

3/7/23

SASTRA DEEMED UNIVERSITY
(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

JULY 2023

Course Code: **MAT102R01**

Course: **ENGINEERING MATHEMATICS - II**

QP No. : **UF019-2**

Duration: **3 hours**

Max. Marks: **100**

PART- A

Answer all the questions

10 x 2 = 20 Marks

1. A particle is travelling along a curve $y^4 + x^3 + x^2 + y^2 - y = 0$. Find the radius of curvature at origin.
2. Write down the working rule to find the extreme values of $f(x, y)$.
3. Determine the curve which touches every member of the family of curves $y = mx + \frac{1}{m}$, m being parameter.
4. By eliminating y , write the equivalent linear differential equation of $x^2 dy + y(x + y) dx = 0$.
5. Obtain the singular solution of $p = \log(px - y)$ where $p = \frac{dy}{dx}$.
6. State Newton's law of cooling.
7. Test whether the following vector field is a divergence free vector field, $\vec{v} = (y^2 - z^2 + 3yz - 2x, 3xz + 2xy, 3xy - 2xz + 2z)$.

8. For a constant vector $\mathbf{a}$, show that $\text{curl}(\mathbf{r} \times \mathbf{a})$ is parallel to $\mathbf{a}$.
9. State Stoke's theorem.
10. What will be the work done by a force $\vec{f} = (y, x)$ which displaces a particle from origin to a point (1,1).

PART- B

Answer all the questions

4 x 15 = 60 Marks

11. (a) Obtain the coordinates of the centre of curvature at any point of the asteroid $x^{2/3} + y^{2/3} = 2^{2/3}$ and hence find its evolute. (8)
- (b) Determine the local extreme values of the function

$$f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$$
 (7)

(OR)

12. (a) Determine the maximum and minimum distance of the point (3,4,12) from the sphere having centre at origin and radius 1. (8)
- (b) Find an envelope of a system of concentric ellipses with their axes along the co-ordinate axes and of constant area. (7)
13. (a) Establish a relation between the two variables x and y satisfying the differential equation $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = e^{-x} \log x$ (8)
- (b) Find the orthogonal trajectory of the family of curves $r^2 = a \sin 2\theta$ where a is parameter. (7)

(OR)

14. (a) The differential equation satisfied by a beam uniformly loaded (W kg/meter), with one end fixed and the second end subjected

to tensile force P is given by $E.I. \frac{d^2 y}{dx^2} = Py - \frac{1}{2} Wx^2$ subject to the conditions $y = 0 = \frac{dy}{dx}$ at $x = 0$. Find the elastic curve for the beam. (8)

(b) Solve $y = 2px + yp^2$, where $p = \frac{dy}{dx}$ (7)

15. (a) Determine the constants m and n such that the surface $mx^2 - 2nyz = (m+4)x$ will be orthogonal to the surface $4x^2y + z^3 = 4$ at the point $(1, -1, 2)$ (8)

(b) Let $\vec{v} = (y \sin z - \sin x)\hat{i} + (x \sin z + 2yz)\hat{j} + (xy \cos z + y^2)\hat{k}$ describes a fluid motion. Is the motion irrotational? If so, find the velocity potential. (7)

(OR)

16. (a) Find the direction in which the directional derivative of $\phi(x, y) = \frac{x^2 + y^2}{xy}$ at $(1, 1)$ is zero and hence find out component of velocity of the vector $\vec{r} = (t^3 + 1)\hat{i} + t^2\hat{j}$ in the same direction at $t = 1$. (8)

(b) Prove that $\nabla^2 f(r) = f''(r) + \frac{2}{r} f'(r)$ where $\vec{r} = (x, y, z)$. (7)

17. Verify Stoke's theorem for $\vec{f} = (x + y)\hat{i} + (2x - z)\hat{j} + (y + z)\hat{k}$ over the surface of a triangular lamina with vertices $(2, 0, 0)$, $(0, 3, 0)$, and $(0, 0, 6)$.

(OR)

18. Verify Gauss divergence theorem over the rectangular parallelepiped $0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$ for

$$\vec{f} = (x^2 - yz)\hat{i} + (y^2 - zx)\hat{j} + (z^2 - xy)\hat{k}.$$

PART- C

Answer the following

1 x 20 = 20 Marks

19. (a) Solve $\frac{d^2 y}{dx^2} + 4\frac{dy}{dx} + 5y = \rho$ where ρ is the radius of curvature of the curve $xy^2 = 8 - x^3$ at the point $(2, 0)$. (10)

(b) Using Green's theorem, find the area of the region in the first quadrant bounded by the curves: $y = x, y = \frac{x}{4}, xy = 1$. (10)
